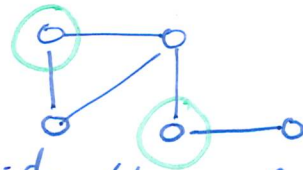


Dominating set.

$U \subseteq V(G)$ is a " " if $\forall u \in V \setminus U$, u has a neighbour in U . e.g.



$$P(v) = \{\text{nbhrs of } v\}, \quad |P(v)| = d_v \quad (\text{degree of } v)$$

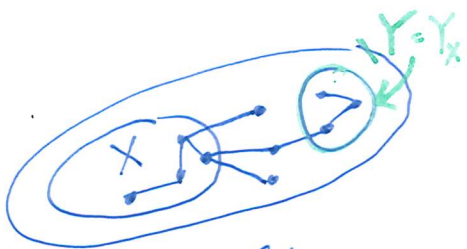
Thm Let $G = (V, E)$, $|V| = n$, minimum deg $\delta \geq 1$

Then G has a dominating set U of size

$$|U| \leq \frac{n}{\delta+1} (1 + \ln(\delta+1)).$$

Pr. Let $p \in [0, 1]$ (choose value later).

For each $v \in V$, indep., place v in X with prob. p .

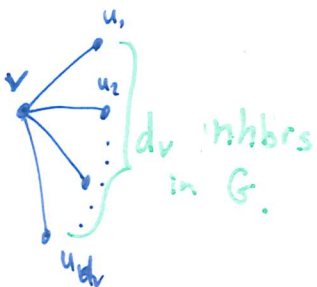


Let $Y = Y_X = \{v \in V \setminus X \text{ s.t. } v \text{ has no neighbour in } X\}$

Calc. $\mathbb{E}[|X|] = \mathbb{E}\left[\sum_{v \in V} \mathbb{1}[v \in X]\right] = np$

$\rightarrow$ fix $v \in V$, $P[v \in Y] = P[v \notin X \text{ and } v's \text{ nbhrs not in } X]$

indep. $\downarrow$
 $= P[v \notin X] \cdot \prod_{u \in P(v)} P[u \notin X]$
 $= (1-p)^{d_v+1}$



$\rightarrow \mathbb{E}[|Y|] = \mathbb{E}\left[\sum_{v \in V} \mathbb{1}[v \in Y]\right] = \sum_{v \in V} (1-p)^{d_v+1} \leq n(1-p)^{\delta+1}$

$\rightarrow \mathbb{E}[|X| + |Y|] \leq np + n(1-p)^{\delta+1}$

$\Rightarrow \exists X' : |X'| + |Y_{X'}| \leq$

$(1-p) \leq e^{-p}$. $\leq np + e^{-p(\delta+1)}$

take $p = \frac{\ln(\delta+1)}{\delta+1}$, $U \stackrel{\text{def}}{=} X' \cup Y_{X'}$.

Thresholds

A+B · 1.3

Let $\mathcal{A}$ be a set of graphs (e.g. with Δ 's as subgraphs)

$$\mathcal{A} = \bigcup_n \mathcal{A}_n \quad \leftarrow \mathcal{A}_n = \{ G \in \mathcal{A} : |V| = n \}$$

Defⁿ

$p^* = p^*(n)$ is a (coarse) threshold for $\mathcal{A}$

if $\bullet \frac{p(n)}{p^*(n)} \rightarrow 0 \Rightarrow \mathbb{P}(G_{n,p} \in \mathcal{A}_n) \xrightarrow{n \rightarrow \infty} 0$

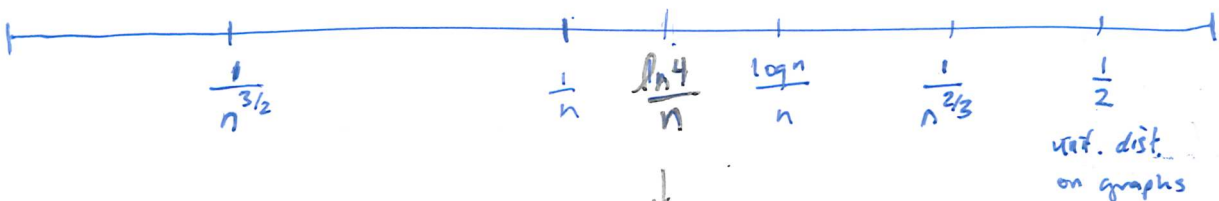
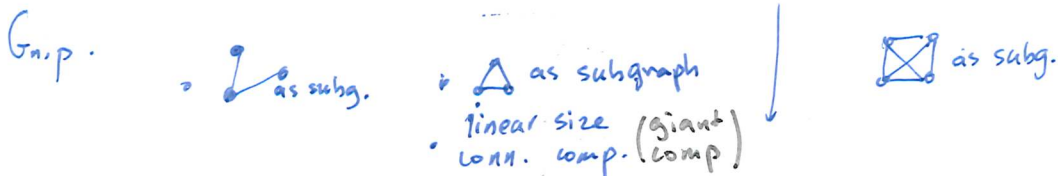
$\bullet$ and $\bullet \frac{p(n)}{p^*(n)} \rightarrow \infty \Rightarrow \mathbb{P}(G_{n,p} \in \mathcal{A}_n) \xrightarrow{n \rightarrow \infty} 1$

" " " sharp threshold

if, $\forall \varepsilon < 1$:

$\bullet p(n) < (1-\varepsilon)p^*(n)$
and
 $\bullet p(n) > (1+\varepsilon)p^*(n)$

Coarse Thresholds: no isol. vert.
connected



Bisection width

sharp threshold

bisection width $\Omega(n)$.
(below b.w. = 0 whp).

hitting time?

$b.w.(G) = \min_{U \subseteq V} e(U, V \setminus U)$
 $|U| = \lfloor n/2 \rfloor$ NP-hard to determine.

(C)

Thm The function $p^*(n) = \frac{1}{n^{2/3}}$ is a threshold
for $G_{n,p}$ containing  as a subgraph.

pf (part i)

Let $\frac{p}{p^*} \rightarrow 0$, i.e. $n^{2/3}p \rightarrow 0$.

Let $G_n \sim G_{n,p}$ w.t.s $P(G_n \text{ contains } \text{K}_4) \rightarrow 0$.

Let $Y_n = \# \text{ of } \text{K}_4 \text{ in } G_{n,p}$.

$$Y_n = \sum_{S \in \binom{[n]}{4}} 1[G_n[S] \cong \text{K}_4]$$

$$\begin{aligned} E[Y_n] &= E\left[\sum_{S \in \binom{[n]}{4}} 1[G_n[S] \cong \text{K}_4] \right] \\ &= \sum_{S \in \binom{[n]}{4}} P[G_n[S] \cong \text{K}_4] = \binom{n}{4} p^6 \\ &\leq \frac{n^4}{4!} p^6 \end{aligned}$$

$$\therefore E[Y_n] \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Note

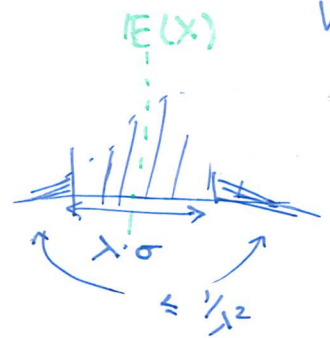
$$\begin{aligned} P(G_n \text{ contains } \text{K}_4) &= P(Y_n > 0) \\ &= \sum_{k=1}^{\infty} P[Y_n = k] \\ &\leq \sum_{k=1}^{\infty} k \cdot P[Y_n = k] \\ &= E[Y_n]. \end{aligned}$$

Thus $P(G_n \text{ contains } \text{K}_4) \rightarrow 0$ as req.

Chebyshev's Inequality

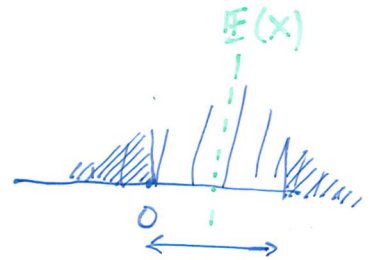
Recall : Let X be a r.v. with mean $\mathbb{E}(X)$, var σ^2 .
 For any $\lambda > 0$, $V(X)$.

$$P(|X - \mu| \geq \lambda \sigma) \leq \frac{1}{\lambda^2}$$



Cor For any r.v. X with $\mathbb{E}(X) > 0$,

$$P(X=0) \leq \frac{\text{Var}(X)}{\mathbb{E}(X)^2}$$



Pf . $P(X=0) \leq P(|X - \mathbb{E}(X)| \geq \mathbb{E}(X))$

$$\leq \frac{\text{Var}(X)}{\mathbb{E}(X)^2}$$

$\mathbb{E}(X) = \lambda \sqrt{\text{Var}(X)}$
 $\lambda = \frac{\mathbb{E}(X)}{\sigma}$
 $= \frac{\mathbb{E}(X)}{\sqrt{V(X)}}$

Pf (part ii).

Let $p : \frac{p}{p^*} \rightarrow \infty$. i.e. $n^{2/3} p \rightarrow \infty$.

" $G_n \sim G_{n,p}$. w.t.s. $P(G_n \text{ contains } \square) \rightarrow 1$.

A_S be event $G[S] \cong \square$

$$Y_n = \sum_{S \in \binom{[n]}{4}} 1[A_S]$$

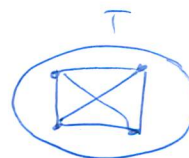
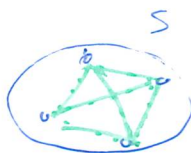
$$\text{Var } Y_n = E[Y_n^2] - E[Y_n]^2$$

$$= E \left[\sum_S 1[A_S] \sum_T 1[A_T] \right] - \left(E \sum_S 1[A_S] \right) \left(E \sum_T 1[A_T] \right)$$

$$= \sum_{S, T} P[A_S \& A_T] - P[A_S] \cdot P[A_T]$$

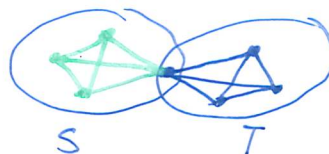
Note

if $S \cap T = \emptyset$



A_S, A_T indep.

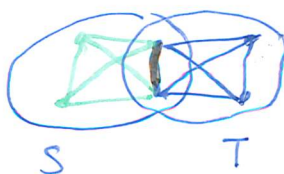
$|S \cap T| = 1$ & $\checkmark$.



$$\Rightarrow \text{Var}(Y_n) = \sum_{\substack{S, T: \\ |S \cap T| \in \{2, 3, 4\}}} P[A_S \& A_T] - P[A_S] \cdot P[A_T].$$

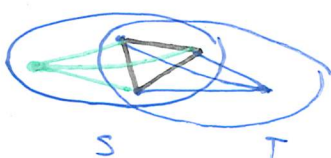
$$\leq \sum_{\substack{S, T: \\ |S \cap T| \in \{2, 3, 4\}}} P[A_S \& A_T] \quad \dots$$

$$|S \cap T| = 2$$



$$P[A_S \& A_T] = p^{11}$$

$$|S \cap T| = 3$$



$$P[A_S \& A_T] = p^9$$

$$|S \cap T| = 4$$

$$P[A_S \& A_T] = P[A_S] = p^6$$

$$\begin{aligned} \therefore \text{Var}(Y_n) &\leq \begin{matrix} \text{pick } S & \text{pick overlap} & \text{pick } T \setminus S \\ \downarrow & \downarrow & \downarrow \end{matrix} \binom{n}{4} \binom{4}{2} \binom{n-4}{2} p^{11} + n^5 p^9 + n^4 p^6 \\ &\leq n^6 p^{11} + n^5 p^9 + n^4 p^6 \end{aligned}$$

by Cor.

$$P(Y_n \neq 0) \geq 1 - \frac{\text{Var}(Y_n)}{\mathbb{E}(Y_n)^2} = 1 - \frac{n^6 p^{11} + n^5 p^9 + n^4 p^6}{\left(\binom{n}{4} p^6\right)^2}$$

$$= 1 - \frac{n^{-2} p^{-1} + n^{-3} p^{-3} + n^{-4} p^{-6}}{\left(\frac{1}{4!}\right)^2 \cdot (1 - \frac{1}{n})(1 - \frac{2}{n})(1 - \frac{3}{n})^2} \xrightarrow{\frac{1}{n^2 p^6} \rightarrow 0} 1$$

□.

divide by
 $n^8 p^{12}$

Ⓐ subgraph containment

$\triangle$ in  is 2. (#labelled copies is more)

notation

$$[n] = \{1, 2, \dots, n\}$$

$$\binom{[n]}{4} = \{S : |S| = 4 \text{ and } S \subseteq [n]\}$$

Ⓑ Erdős-Rényi / Binomial random graph

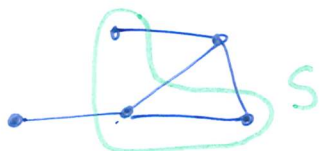
Defⁿ $G_{n,p}$. Sample $G_n \sim G_{n,p}$ by taking $V(G_n) = [n]$

and let $uv \in E(G_n)$ with prob p indep for each pair of vertices (u, v) , $u, v \in [n]$.

Ⓒ $G[S]$: graph restricted to $S \subseteq V(G)$.

Defⁿ $G[S]$ is the graph obtained by deleting vertices in $V \setminus S$ and any edges incident to these vertices.

e.g. G



$G[S]$

